ECE-350 Fall 2026

Interactive Embedded Systems Design Project

Description:

Students work in 2-person engineering teams to design a prototype an interactive embedded system that offers value to a stakeholder. Project presentations occur in the form of a showcase event at The MIX.

Project Requirements:

- Teams must be formed in Canvas using the self-sign up feature. Due: 10/21
- Projects must incorporate the Raspberry Pi and various other components from the lab kit.
- Additional components may be used (and is encouraged!)
- All projects must incorporate a feature that provides some form of interaction with the user
- A proposal needs to be submitted to Canvas (Due: 10/21) describing the project with the following value statement:
  Our (project) helps _____by _____so they can _____
- Teams may opt-in to work on one of the 4 hydroponics projects posted on Canvas
- Teams may also propose projects for the Sunfounder robotic dog
- Teams are encouraged to design and build enclosures or other custom parts using 3D printers, cardboard, or other materials to enhance their designs.
- Use of code:
  • Aside from writing your own code, you may use code provided by Adeept, code provided as part of gpiozero documentation, or code supplied by the professor.
  • You may also use external libraries.
  • Any code that is not your own **must be clearly indicated** in your code submission through python comments. This includes:
    - code from Adeept,
    - code from gpiozero documentation
    - code provided by the professor,
    - any external libraries used.

  •Your comments must also include a link to the original source for any external code or library.
  • You may also use Generative AI (GenAI) to generate code; however, any AI generated code must likewise be clearly identified in comments as not your own.
- **Use of GenAI tools is encouraged** and may be used for project brainstorming, ideation, enhancements and code generation. If you are using GenAI for code generation you will need to indicate in your code submission through comments in the code.
- **All code files need to be submitted to Canvas by end of day the project is presented**
- You will also need to indicate what aspects of your project benefited from GenAI through reflective assignments.
- Students will vote for the best projects once the presentations are over. Winners will be announced on final exam day.

- Project Presentations:
    - Group 1: Wednesday 12/2
    - Group 2: Monday 12/7

Category	**Points**
Proposal (with value statement)	**2 points**
Overall Design (functionality, complexity, craftsmanship, enhancements, aesthetics, cable management, ease of interaction, creativity)	**10 points**
Project demonstration (working prototype)	**10 points**
Well-commented Python code	**4 points**
Total	**26 points**